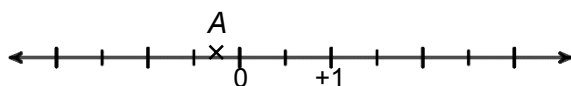


Section A: Multiple-choice questions. Choose the best answer for each question. (30 marks)

1. Refer to the horizontal number line below.



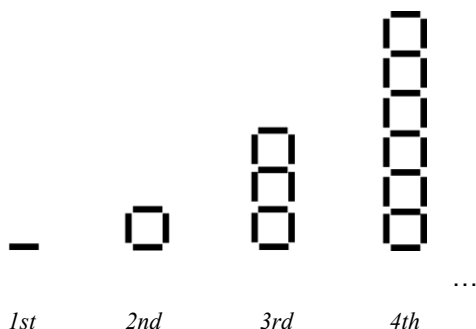
Which of the following is a possible value of A ?

- A. $+0.3$
B. -0.5
C. $-\frac{1}{4}$
D. $-\frac{3}{5}$
2. If a, b, c, d, e, f, g and h are all positive numbers, is the value of the following expression positive or negative?

$$-(-a)(b)\left(\frac{-c}{d}\right)(e)\left(\frac{1}{f}\right) \div (-g)(-h)$$

- A. Positive
B. Negative
C. Neither positive nor negative
D. Cannot be determined
3. John is $3d$ years old, and his grandfather is $4d$ times as old as him. Find the age of John's grandfather.
- A. $7d$
B. $24d$
C. $7d^2$
D. $12d^2$
4. Given the formula $H = a^2 + 2b$, find the value of H when $a = -5$ and $b = 3$.
- A. -19
B. -4
C. 16
D. 31

5. The following shows a sequence of figures formed by sticks. The 1st figure consists of 1 stick. For any positive integer n , the $(n + 1)$ th figure is formed by adding $3n$ sticks to the n th figure.



Find the number of sticks in the 7th figure.

- A. 31
 B. 46
C. 64
 D. 85
6. Arrange the following angles in ascending order of their sizes.

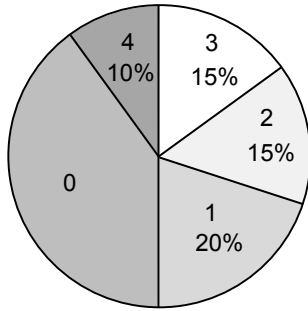
$$\angle A = \text{acute angle} \qquad \qquad \qquad \angle B = \frac{2}{3} \text{ reflex angle}$$

$$\angle C = \frac{3}{5} \text{ straight angle} \qquad \qquad \qquad \angle D = \frac{3}{4} \text{ round angle}$$

- A. $\angle A < \angle B < \angle C < \angle D$
B. $\angle A < \angle C < \angle B < \angle D$
 C. $\angle D < \angle B < \angle C < \angle A$
 D. $\angle D < \angle C < \angle B < \angle A$

7. The pie chart below shows the numbers of pets that 80 households have.

Numbers of pets that 80 households have

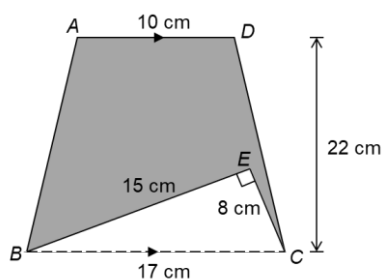


How many households do NOT have any pets?

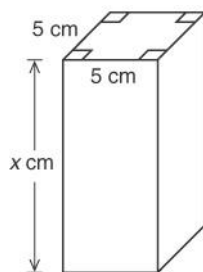
- A. 16
 - B. 24
 - C. 32**
 - D. 40
8. There are 125 pieces of biscuits in a box. 30 of them are orange-flavoured. Find the percentage of orange-flavoured biscuits in the box.
- A. 24%**
 - B. 25%
 - C. 30%
 - D. 76%
9. The price of a necklace is changed from \$200 to \$150. The percentage change in price is
- A. -25%.**
 - B. -75%.
 - C. +25%.
 - D. +75%.
10. A TV set is sold for \$2380 at a loss of 15%. Find the cost price.
- A. \$2023
 - B. \$2070
 - C. \$2737
 - D. \$2800**

11. $0.070\ 425 =$
- A. 0 (cor. to 1 sig. fig.)
 - B. 0.07 (cor. to 2 d.p.)**
 - C. 0.0704 (cor. to 4 sig. fig.)
 - D. 0.070 42 (cor. to 4 d.p.)
12. Estimate the value of $243.7 + 37.45 + 4.743$ by rounding down each number to the nearest integer.
- A. 283
 - B. 284**
 - C. 285
 - D. 286

13. Find the area of the shaded region in the figure.

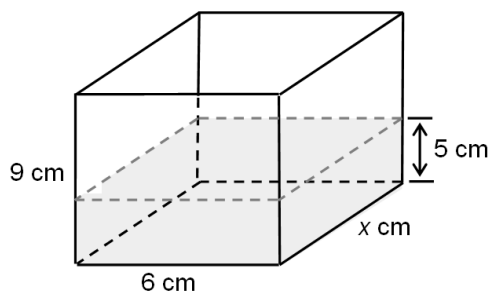


- A. 237 cm^2**
 - B. 269 cm^2
 - C. 297 cm^2
 - D. 357 cm^2
14. In the figure, the volume of a cuboid of height $x\text{ cm}$ is 280 cm^3 . Find the value of x .



- A. 6
- B. 9
- C. 11.2**
- D. 14

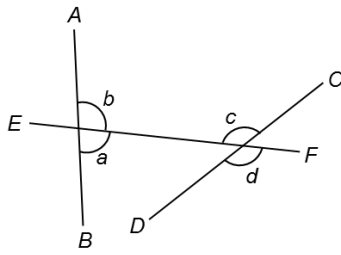
15. The figure below shows a rectangular tank filled with water. After putting eight cubes, each of side 3 cm, into the tank, the tank becomes fully-filled with water and no water overflows. If all the cubes are totally submerged, find the value of x .



- A. 6
 B. 7
 C. 8
D. 9
16. If the degree of the monomial $2p^2q^3r^m$ is 12, find the value of m .
 A. 1
 B. 2
 C. 5
D. 7
17. Which of the following is NOT a polynomial?

- A. $\frac{2}{3}x - 1$
B. $\frac{x}{y} + 5$
 C. -7
 D. $\frac{x^2}{4} - \frac{y^2}{3}$

18. In the figure, AB , CD and EF are straight lines. Which of the following is a pair of alternate angles?



- A.** a and c
- B.** a and d
- C.** b and c
- D.** c and d
19. PQ is a straight line parallel to the y -axis. If the coordinates of P are $(-7, 4)$, which of the following can be the coordinates of Q ?
- A.** $(-7, 7)$
- B.** $(4, 7)$
- C.** $(4, -7)$
- D.** $(7, 4)$
20. $P(-5, 8)$ is a point on a rectangular coordinate plane. Which of the following transformations gives a different image from the others?
- A.** P is translated to the right by 10 units.
- B.** P is reflected about the y -axis.
- C.** P is rotated clockwise about the origin through 90° .
- D.** P is reflected about the x -axis and then rotated clockwise about the origin through 180° .

***** End of Section A *****